CODECHEF

The CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest, is now in print!

Hello and welcome to the CodeChef Puzzle Challenge sponsored by LINUX For You and Directi! Solve this month's puzzle, and you could be one of the three lucky people to win a cash prize of Rs 1,000 each!

The June edition's puzzle:

A boy lives on the third floor of a building at a height of 12 metres from the ground. He spots a hot air balloon in the sky at an inclination of 45°. At the same time, a man standing at the bottom of the building spots it at an inclination of 60°. The boy runs to his mother to tell her about it. After 10 seconds, his mother comes out to have a look at the balloon. The angle of inclination of the balloon remains the same, i.e., 45°. Can you find the altitude of the balloon from the ground when the mother sees it? Consider the horizontal speed of balloon to be 3 m/sec. Also find the vertical speed of the balloon.

The Solution:

Using figure (1)

At position Q, let PO = X m

Since the angle of inclination is 45° from A, and tan 45°=1, the vertical distance from the 3rd floor is the same, i.e., QT=X m.

Hence, from the ground, PQ = (X+12) m … (1)

Since the angle of inclination is 60° from O, and tan 60°= √3, then,

PQ/PO = √3

PO =PQ/ √3

√3X= X+12

X= 12/(√3-1)

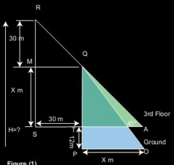

Figure (1)

X= 16.39… (2) using √3= 1.732

Now after the mother comes out to look at the balloon, it has travelled to position R, yet the angle of inclination remains the same, i.e., 45°. Since tan 45°=1, the vertical distance is equal to the horizontal distance, i.e., RS=SA … (3)

SA = (X+30) m … (4)

Now, H = (X+30) + 12 … (5) using (3) and (4)

H= 16.39+30+12 using (2). Hence, the height of the balloon = 58.39 m.

The vertical speed of the balloon = 3 m/sec since it remained at the same angle from A.

[Note: The '√ ' sign is used for square root.]

And the winners this month are:

- Gunjan Gupta
- Priyadarshini Parida
- B N Singh

Here's your CodeChef *Puzzle of the Month*:

Our chef planned to open a new restaurant. He went to the market and bought some plates and bowls. He hired a person to take the plates and bowls to the restaurant. Unfortunately, the man broke some of them on the way and managed to only deliver exactly 2/3rd of the bowls and 3/4th of the plates safely. If an equal number of bowls and plates were found to be intact, what fraction of the total items got damaged?

Looking for a little extra cash?

Solve this month's CodeChef Challenge puzzle and send in your answer to *codechef@efyindia.com* or *lfy@codechef.com* by July 15, 2012. Three lucky winners will get some awesome prizes both in cash and merchandise! You can also participate in the contest by visiting the online space for the article at *https://www.facebook.com/LinuxForYou*. END

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the July Challenge that takes place from the 1st to the 11th of every month to win cash prizes of up to Rs 20,000. Also keep visiting CodeChef and participate in the multiple programming contests that take place on the website throughout the month!